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# Paleomagnetism of Middle Triassic redbeds from Hubei and northwestern Hunan provinces, South China

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**Abstract**

Paleomagnetic investigations have been conducted on redbeds from the Middle Triassic Badong Formation at Badong, Hubei and Sangzhi, Hunan as well as from the coeval Puxi Formation from Puxi, southeastern Hubei province, South China. The characteristic remanent magnetization (ChRM) determined from Badong and Sangzhi passes the fold test with dual polarity. The tilt-corrected ChRM direction from Puxi is congruent with those from Badong and Sangzhi but is based on a smaller number of samples collected from a monocline. These ChRM directions are believed to be primary. The data indicates that Badong has been rotated clockwise by $13.6 \pm 8.5°$ relative to Sangzhi, which, in turn, has possibly been rotated in the same sense with respect to other parts of the Yangtze Block (YB) except eastern Sichuan based on the bend of the sampled fold axis and the Lower Triassic paleomagnetic data published for the YB. The Lower and Middle Triassic paleomagnetic data together appear to indicate that eastern Sichuan and the border area between Sichuan, Guizhou, Hubei and Hunan provinces have been affected by differential rotations, probably due to oroclinal bending. The paleomagnetic data obtained from this study further constrain the timing of the final suturing of the YB with the North China Block (NCB) to be post-Middle Triassic.

*Keywords:* paleomagnetism; Middle Triassic; Hunan China; Hubei China; Yangtze Platform; oroclines

## 1. Introduction

Paleomagnetic data have constrained the final suturing of the Yangtze Block with the North China Block to be sometime between Early Triassic and Cretaceous times [1–7]. To further constrain the timing, reliable paleomagnetic results are needed from rocks with ages spanning the Middle Triassic through the Jurassic from both blocks. However, reliable Middle to Late Triassic paleomagnetic results are still lacking for the YB. The situation is best illustrated by the fact that Enkin et al. [7] reviewed the Middle and Late Triassic paleomagnetic poles for the YB and considered only one of them to be reliable. This pole (51.0°N, 189.5°E, $A_{95} = 6.7°$) is a recalculated average of four poles reported by Zhu et al. [8] from southwestern Sichuan province from redbeds of the Upper Triassic Bingnan Formation and from coal-bearing Baiguowan and Xujiahe Formations. We made a much more extensive sampling of the Bingnan Formation at the same locality in 1987; however, few samples from our collection

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yielded stable paleomagnetic directions during thermal cleaning. Moreover, most samples Zhu et al. collected from the coal-bearing Baiguowan and Xujiahe Formations were weakly magnetized, so it is difficult to be sure if the astatic and Schonstedt spinner magnetometers they used would be sensitive enough to adequately measure the magnetization of these rocks. Thus, the reliability of even this pole can be questioned.

The need for high-quality Middle Triassic to Jurassic paleomagnetic results for the YB is further highlighted by recent studies conducted by Wang and Van der Voo [9] and Wang et al. [10] on the Paleozoic rocks from Nanjing and Middle Triassic rocks from southern Guizhou province. They identified a 'folding-associated' remagnetization of probably Triassic to Jurassic age from these studies and proposed a new Mesozoic loop to be added to the apparent polar wander path (APWP) for the YB [10]. Apparently, rocks which may preserve primary magnetization must be sought and investigated in order to confirm or refute this proposed Mesozoic loop in the APWP for the YB based purely on secondary magnetizations.

Thus, studies of the paleomagnetism of the Middle Triassic redbeds from Hubei and northwestern Hunan provinces were initiated.

## 2. Geological setting

It is known that some Middle Triassic rock formations in the middle and lower reaches of the Yangtze River contain redbeds or redbeds interbedded with marine carbonates. They include the Badong Formation in the border area between Sichuan, Hubei and Hunan provinces, the Puxi Formation in southeastern Hubei province, the Tongtoujian Formation in southern Anhui province, and the Huangmaqing Formation in southern Jiangsu province [11–14]. These and their companion Middle Triassic formations conformably overlie Lower Triassic shallow marine carbonates and conformably or disconformably underlie Late Triassic continental coal-bearing series, both the underlying and overlying rocks having very good fossil control, and they typically yield a bivalve assemblage *Eumorphotis* (*Asoella*) *illyrica–Costatoria radiata–Entolium discites*, which is called the Badong fauna by some Chinese paleontologists [13].

At the type section in the Yangtze Gorge area in western Hubei, the Badong Formation totals 1148 m in thickness and is divisible into five members [11]. Member 1 is composed of greenish-gray calcareous shale interbedded with thin-bedded limestone in the upper part and gray limestone and dolomitic limestone in the lower part, with the latter transitional from the underlying Jialingjiang Formation of Lower Triassic age. Member 2 consists primarily of purplish-red siltstone and mudstone. Member 3 is characterized by gray limestone with intercalated calcareous shale. Member 4 is again composed of purplish-red calcareous siltstone and silty mudstone with subordinate sandstone. Member 5 is made up of dark gray dolostone with intercalations of silty shale and is conformably overlain by Late Triassic Shazhenxi Formation. Members 1, 3 and 5 are believed to be platform carbonate deposits and yield abundant marine fossils, such as bivalve assemblages *Costatoria goldfussi mansuyi–Eumorphotis* (*Asoella*) *illyrica* in Member 1 and *Plagiostoma striatum–Placunopsis plana* in Member 3, a foraminiferal assemblage *Glomospira nanzhangensis–Quasipadangia baotaheensis* and ammonites *Progonoceratites* in Member 3. Members 2 and 4 are regarded as tidal flat sediments and contain fewer marine fossils. Still, an ostracod assemblage *Darwinula lauta–Darwinuloides graphicus*, a charophyte assemblage *Porocharo–Vladimiriella*, and a conchostraca assemblage represented by *Xiangxiella* are present in Member 2 and a charophyte assemblage *Stellatochara–Stenochara* in Member 4. A rich flora represented by *Pleuromeia–Annalepis* has been recently found from Member 2 [15]. Based upon these fossil records, Members 1 through 3 have been assigned an Anisian age and Members 3 and 4 a Ladinian age [11,15].

In southeastern Hubei, the coeval rocks are divided into two formations: the Lushuihe and Puxi Formations [12]. At the type section on the outskirts of Puxi City, the Lushuihe Formation is 87 m thick, and conformably overlies the Lower Triassic Guanyinshan Formation, an equivalent of the Jialingjiang Formation in western Hubei and Sichuan provinces, and is composed of gray limestone and dolostone in the lower part and greenish-gray siltstone and silty mudstone in the upper part. The

overlying Puxi Formation is 378 m in thickness and comprises purplish-red silty mudstone and calcareous siltstone with subordinate sandstone and conglomerate intercalations. The fossils are similar to those of the Badong Formation, including bivalves *Eumorphotis* (*Asoella*) *subillyrica*, *E.* (*A.*) *illyrica*, etc., and plant *Annalepis* sp. from the Lushuihe Formation [12]. A rich sporo-pollen assemblage indicative of Middle Triassic age was also reported from the Lushuihe Formation [11].

The samples for this study were collected from the Badong Formation at two localities with one being the type section in western Hubei and another in northwestern Hunan, and also from the Puxi Formation at the type locality in southeastern Hubei.

Hubei and northwestern Hunan provinces involve

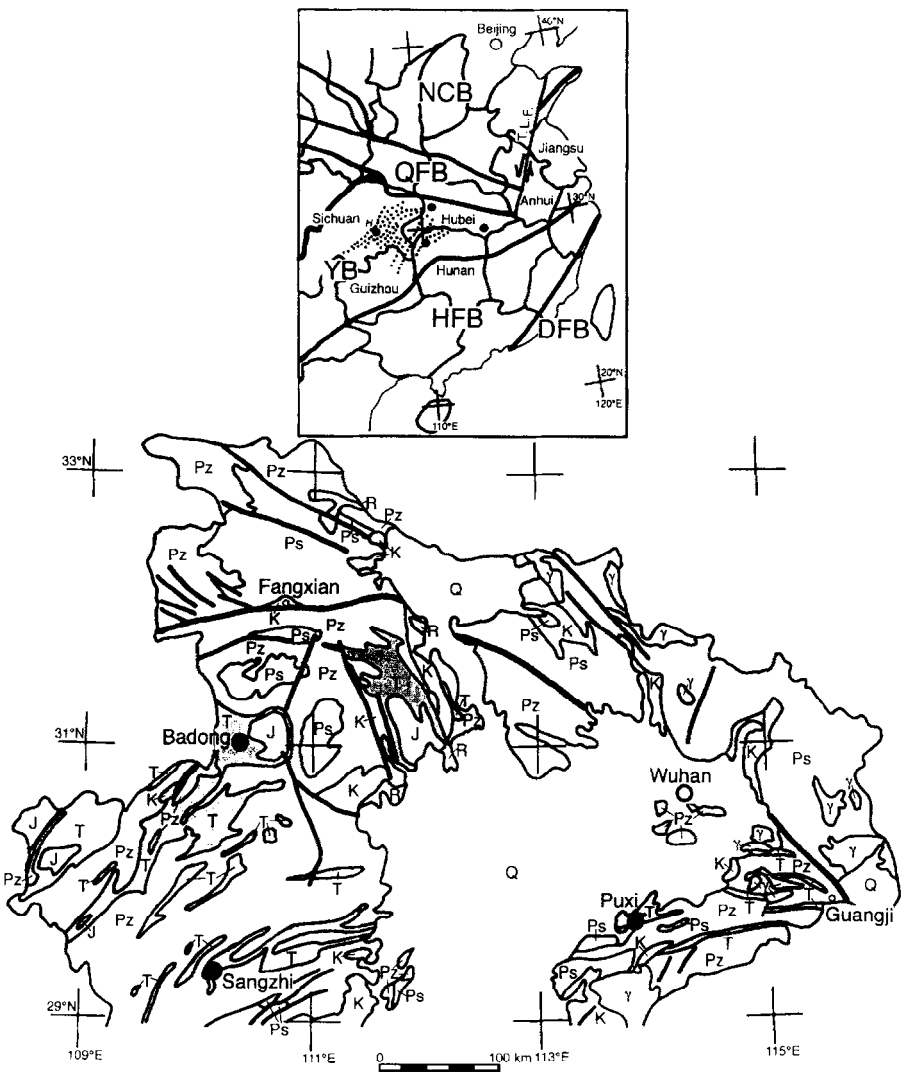



Fig. 1. Sketch geological map of Hubei and northern Hunan provinces. Dots denote the sampled localities of this study. Strata: *Ps* = pre-Sinian; *Pz* = Sinian–Permian; *T* = Triassic; *J* = Jurassic; *K* = Cretaceous; *R* = Tertiary; *Q* = Quaternary. Triassic outcroppings are shaded. Granitic intrusions are indicated with γ. Heavy lines are major faults. The dotted lines in the inset show the fold axes in eastern Sichuan and the border area between Sichuan, Guizhou, Hubei and Hunan provinces. The dot with *H* denotes the sections at Hechuan in eastern Sichuan investigated by Steiner et al. [5]. Abbreviations for the major Chinese tectonic units: *NCB* = North China Block; *YB* = Yangtze Block; *QFB* = Qinling Fold Belt; *HFB* = Huanan Fold Belt: *DFB* = Dongnanyanhai Fold Belt. *T.L.F.* = Tan-Lu Fault.

two major tectonic units: the Qinling Fold Belt and the Yangtze Block, the two being separated by the Fangxian–Guangji Fault (Fig. 1.). The Qinling Fold Belt within Hubei province is part of its southernmost zone composed of Sinian to Triassic passive-margin sediments on basement rocks of the YB which were folded and overthrust southward onto the YB when the YB collided with NCB [16,17]. Recent discoveries of coesite and coesite pseudomorphs in omphacite and garnet as well as diamond in garnet in eclogite from the eastern extension of this zone (Dabieshan) in adjacent Anhui province indicate that the basement of the YB had been thrust into the mantle to a depth of more than 100 km and subsequently rapidly exhumed [18,19]. It should be pointed out that the Yangtze Block has sometimes been improperly equated to the South China Block (SCB) in the past ten years or more. The fact is that mainland South China has long been recognized to be composed of three major tectonic units: the YB, Huanan (South China) Fold Belt and Dongnanyanhai (Southeastern Coast) Fold Belt, with the latter two accreted onto the YB during the Paleozoic [20]. In recent years, Hsu and his collaborators postulated that the South China Fold Belt, which they termed as the Huanan Block, collided with the YB during the Triassic instead of the end-Silurian, and identified the Precambrian Banxi Group basically as a collision melange, part of which is exposed in central Hunan province [21,22]. Their arguments are, however, disputed by some scientists [23]. Nonetheless, our sampling sites are located within the undisputed part of the YB (see Fig. 1).

It is worth mentioning that there are some prominent folds in the sedimentary cover strata of the YB in eastern Sichuan and the border area between Sichuan, Guizhou, Hubei and Hunan provinces. These folds trend northeast to north-northeast along most of their length but bend into a roughly east–west direction when they approach the Qinling Fold Belt, and are characterized by box-shaped cross-section (in the border area between Sichuan, Guizhou, Hubei and Hunan) or narrow and tight anticlines separated by broad and flat synclines (in eastern Sichuan). They constitute the Bamianshan and Chuandong (eastern Sichuan) arcs, respectively, as termed by some Chinese geologists [24]. The folding took place before the Cretaceous as evidenced by an unconformity between the Cretaceous and older rocks [12,24]. The samples of the Badong Formation from the two localities were collected from two of these folds.

## 3. Laboratory techniques

Because all the samples are from redbeds, stepwise thermal demagnetization was used throughout the investigation. The samples were demagnetized in a Schonstedt oven and measured in a 2G cryogenic magnetometer. All the equipment is housed in a magnetically shielded room. Orthogonal projections were employed to examine the demagnetization behavior of the samples [25]. Paleomagnetic directions were calculated using principal component analysis technique [26]. Site and formation means were computed using Fisherian statistics [27].

## 4. Sampling and results

### *4.1. Badong Formation from Badong, western Hubei (31.0°N, 110.4°E)*

The type section for Badong Formation is outside the town of Badong county located on the western flank of the Zigui syncline, a basin-like syncline with the Jurassic rocks in the core and Triassic rocks cropping out along the flank (Fig. 1). Whereas the Badong Formation on the eastern flank is thin and nearly pinches out, it is much thicker (see earlier description of the type section), and refolded on the western flank. These second-order folds trend east–west. Our samples from this locality were collected from Member 2 from two steep-dipping limbs of a box-shaped syncline. At the southern limb, the rocks consisting of Members 1 and 2 dip north at 48–70° and the exposures are very fresh and almost continuous along a ravine on the southern bank of the Yangtze River. Therefore, cores were drilled, using a gasoline-powered portable rock drill and oriented with a magnetic compass, with a sampling interval of about 5 m along the section so that a magnetostratigraphy could be obtained. A total of 61 cores were collected nearly covering the total thickness of 355 m of Member 2 at the section. Greenish intercalations were intentionally avoided. At the northern

limb, Member 2 crops out about 2 km north of downtown Badong along the road from Badong to Xingshan, where the beds dip south at 69–89°. Seven sites were drilled with 5–6 cores per site.

The intensities of the natural remanent magnetization (NRM) of the samples from the southern limb vary from 1.6 to 13.9 mA/m with the majority in the range 4–9 mA/m. During thermal demagnetization, a component, designated as component A herein, was removed from most of the samples below 650°C (Fig. 2a–d). Component A accounts for 40–80% of the initial NRM intensities, and is northeast or north-northeast directed with moderate to steep-down inclinations in in-situ coordinates. Above 660°C, a ChRM component, designated as component B, was resolved, and two polarities are present. At the sampled section, Member 2 is dominantly of normal polarity with a reversed magnetozone near the base. In in-situ coordinates, the N-polarity directions are easterly or east-southeasterly with moderate and downward inclinations (Fig. 2b–d) and the R-polarity directions are westerly or northwesterly with shallow and upward inclinations (Fig. 2a), none of them resembling the Earth's present-day field direction. For statistical reasons, we grouped the adjacent samples into sites, and ten sites were thus obtained with 6–7 samples in a site.

Samples from site 1 from the northern limb, which were drilled from right below the limestones of Member 3, are relatively weakly magnetized with NRM intensities ranging 1–3 mA/m and did not yield stable ChRM directions during thermal treatment, and are thus excluded from further analysis. The NRM intensities of the samples from the remaining 6 sites are comparable to that of the majority of the samples from the southern limb. Similarly,

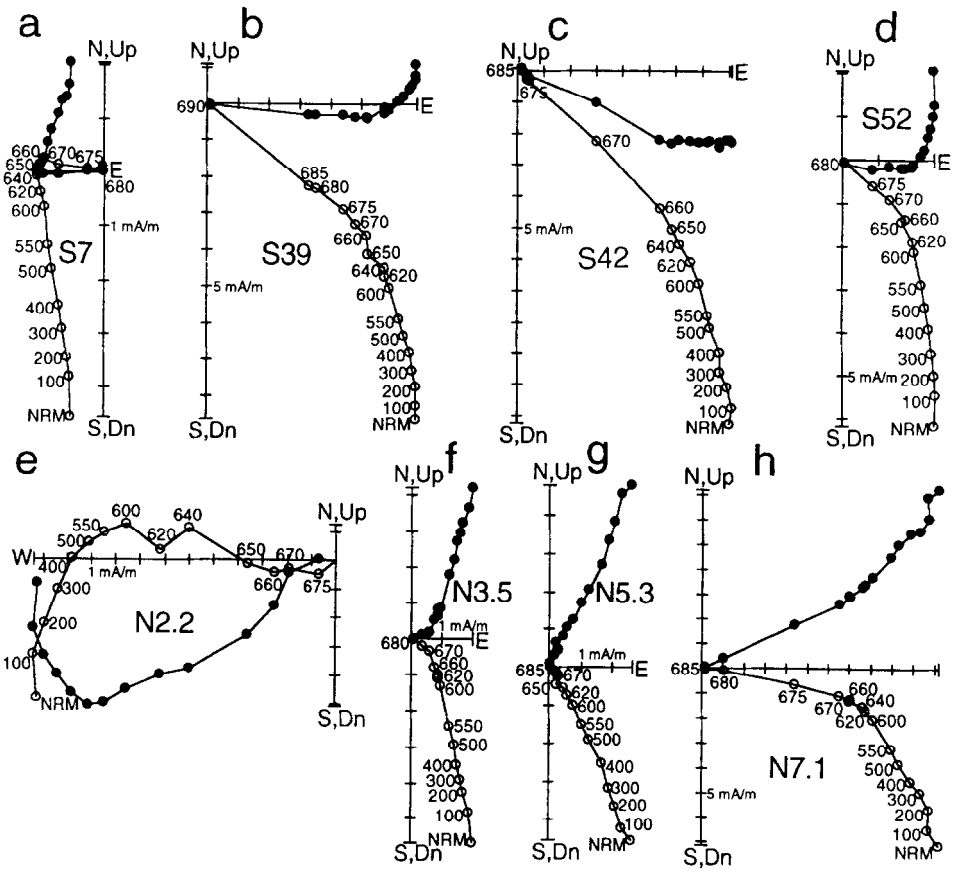



Fig. 2. Representative orthogonal plots of thermal demagnetization (in in-situ coordinates) for the Badong Formation from Badong, Hubei. Dots and circles are plotted on horizontal and vertical planes, respectively. Treatment levels are in °C.

they possess two components: A and B; component A, however, predominates in most samples, amounting to 85–90% of the initial intensities, and persists at up to 660°C or even higher temperatures (Fig. 2f). In some cases, component A entirely prevails and no component B was resolved (Fig. 2g). Only in a few samples does component B account for an appreciable proportion of the NRM intensities (Fig. 2h) as do most samples from the southern limb. Again, two polarities are present in component B, with 3 samples from site 2 reversed (Fig. 2e) and all other samples normal (Fig. 2f,h).

Whilst the mean in-situ directions of component A for the 16 sites from both limbs are reasonably well grouped, those of component B are quite scattered (Fig. 3 left). Upon restoring the two limbs to horizontal by a simple rotation about the strike, the site mean directions of component A are split into two groups ($\kappa_s/\kappa_g = 0.05$), whereas the grouping of those of component B is improved ($\kappa_s/\kappa_g = 1.69$) but not significantly ($\kappa_s/\kappa_g < 1.84$, the critical value for a fold test to be positive at the 95% confidence, see [28]) (Fig. 3 middle). Since these east–west-trending folds are developed on the western flank of the Zigui syncline, a plunge is expected to exist. To check and determine the plunge of the sampled fold, we employed the iterative method [29] to determine the mean intersection of the bedding planes measured from the two limbs. The plunge was determined to be 87.6° east at 25.7° (Fig. 4), consistent with the general attitude of the western flank of the Zigui syncline. We then unplunged the fold axis followed by a rotation about the bedding strike. The procedures used basically followed Schmidt [30] with the following modification: normals to the bedding planes were regarded as vectors which were rotated and recalculated as were the magnetic vectors during the unplunging so that the simple rotation about the strike could then be universally applied irrespective of which limb of the fold the samples are from. The site mean directions thus obtained are listed in Table 1 and plotted in Fig. 3 right. It is clear that component A still does not pass the fold test with $\kappa_s/\kappa_g =$ 0.52, while component B does pass the fold test at

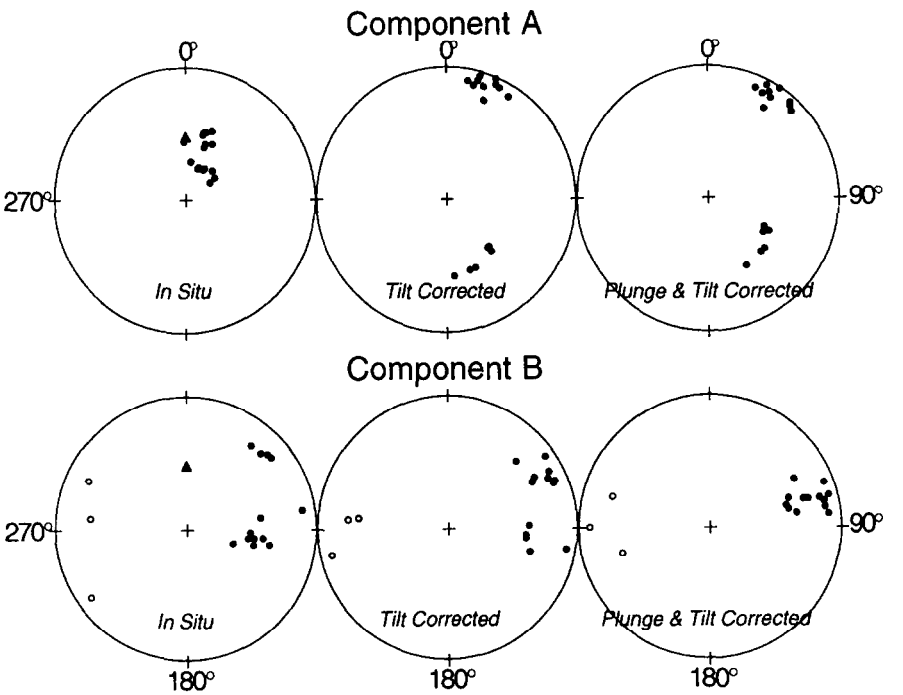



Fig. 3. Equal-area projections of site mean directions of components A and B for the Badong Formation from Badong, Hubei. Dots and circles represent directions plotted onto the lower and upper hemispheres, respectively. Black triangle denotes the present dipole field direction at the sampled locality.

99% confidence level ($\kappa_s/\kappa_g = 3.24 > 2.38$, see [28]). The plunge and tilt corrected mean direction for the Badong Formation from this locality is $D/I = 77.6°/23.0°$ with $\alpha_{95} = 6.8°$, which yields a pole at 16.6°N/195.8°E with $A_{95} = 5.5°$.

### 4.2. Badong Formation from Sangzhi, northwestern Hunan (29.4°N, 110.2°E)

The Badong Formatopm was also sampled in the vicinity of the town of Sangzhi in northwestern

Table 1
Site and formation statistics for the Badong Formation from Badong (31.0°N, 110.4°E)

| Site | | Bedding | | n/N | In situ | Plunge and tilt corrected | | | | Pole position | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | D (°) | I (°) | D (°) | I (°) | $\alpha_{95}$ | $\kappa_s$ | Latitude (°N) | Longitude (°E) |
| S1 [a] | 281/50 | B | 5/6 | 296.5 | −18.7 | 286.9 | −23.9 | 25.1 | 10.3 | 7.6 | 0.9 |
| | | A | 6/6 | 20.8 | 55.2 | 28.3 | 4.2 | 5.8 | 135.1 | 72.0 | 179.6 |
| S2 [a] | 284/56 | B | 5/6 | 276.8 | −26.9 | 270.2 | −10.0 | 19.9 | 15.8 | −2.4 | 16.0 |
| | | A | 6/6 | 8.0 | 66.1 | 24.0 | 10.6 | 11.8 | 33.1 | 71.5 | 127.3 |
| S3 | 292/55 | B | 5/6 | 97.5 | 47.0 | 80.1 | 12.7 | 15.6 | 25.0 | 11.8 | 199.9 |
| | | A | 6/6 | 43.8 | 65.8 | 44.5 | 10.4 | 9.5 | 50.2 | 52.8 | 160.3 |
| S4 | 291/53 | B | 6/6 | 93.3 | 49.8 | 75.3 | 15.9 | 11.5 | 34.7 | 16.7 | 200.9 |
| | | A | 6/6 | 21.0 | 53.1 | 33.8 | 1.5 | 3.3 | 407.1 | 72.1 | 186.3 |
| S5 | 279/58 | B | 4/6 | 103.3 | 46.1 | 74.2 | 24.0 | 14.6 | 40.4 | 19.9 | 197.4 |
| | | A | 6/6 | 31.7 | 67.6 | 32.6 | 11.8 | 11.0 | 37.8 | 59.5 | 151.5 |
| S6 | 284/58 | B | 5/6 | 107.4 | 59.7 | 60.7 | 28.3 | 7.6 | 101.7 | 32.6 | 201.8 |
| | | A | 6/6 | 23.7 | 68.6 | 30.5 | 8.9 | 13.5 | 25.8 | 62.8 | 143.3 |
| S7 | 279/63 | B | 6/6 | 96.6 | 41.1 | 77.4 | 12.5 | 8.1 | 69.5 | 14.0 | 201.4 |
| | | A | 6/6 | 54.0 | 67.3 | 42.3 | 7.3 | 8.4 | 64.1 | 45.9 | 158.7 |
| S8 | 286/70 | B | 5/6 | 100.5 | 35.7 | 84.0 | 11.1 | 11.1 | 48.8 | 8.0 | 198.7 |
| | | A | 6/6 | 55.4 | 71.2 | 41.5 | 5.9 | 9.2 | 54.1 | 44.4 | 150.8 |
| S9 | 281/58 | B | 5/6 | 81.2 | 42.1 | 69.1 | 8.6 | 5.9 | 171.6 | 20.1 | 207.7 |
| | | A | 6/6 | 22.7 | 68.9 | 28.0 | 12.8 | 13.5 | 25.4 | 63.0 | 141.7 |
| S10 | 278/48 | B | 5/7 | 98.6 | 50.4 | 73.2 | 27.3 | 22.9 | 12.2 | 21.6 | 196.1 |
| | | A | 6/7 | 31.9 | 68.1 | 32.3 | 21.6 | 10.3 | 43.4 | 59.0 | 150.4 |
| N2 [a] | 86/85 | B | 6/6 | 234.7 | −11.6 | 253.1 | −31.6 | 12.7 | 28.9 | −22.9 | 13.6 |
| | | A | 6/6 | 358.9 | 54.1 | 150.3 | 40.9 | 13.4 | 25.8 | 86.3 | 96.4 |
| N3 | 89/89 | B | 4/5 | 80.9 | 11.5 | 75.1 | 8.1 | 22.4 | 17.8 | 14.9 | 204.6 |
| | | A | 5/5 | 17.9 | 45.9 | 135.9 | 42.2 | 4.6 | 277.9 | 74.0 | 209.2 |
| N4 | 86/88 | B | 5/5 | 50.8 | 17.4 | 81.1 | 34.1 | 12.4 | 39.3 | 16.9 | 188.4 |
| | | A | 5/5 | 16.4 | 46.0 | 132.1 | 42.5 | 4.9 | 245.3 | 75.2 | 210.5 |
| N5 | 82/76 | B | 5/6 | 45.2 | 20.6 | 77.5 | 39.0 | 9.2 | 70.4 | 21.4 | 186.8 |
| | | A | 6/6 | 22.6 | 44.1 | 117.4 | 50.2 | 3.9 | 298.0 | 69.5 | 209.3 |
| N6 | 77/78 | B | 5/6 | 38.6 | 21.1 | 74.7 | 39.9 | 17.5 | 20.2 | 24.0 | 187.4 |
| | | A | 6/6 | 16.1 | 47.8 | 118.6 | 46.8 | 3.0 | 513.8 | 75.9 | 205.0 |
| N7 | 82/69 | B | 5/6 | 48.1 | 17.2 | 70.6 | 37.0 | 18.7 | 17.7 | 26.6 | 191.3 |
| | | A | 6/6 | 26.1 | 51.4 | 121.8 | 49.7 | 6.8 | 96.7 | 67.8 | 190.8 |
| Formation Mean | | B | 16/17 | 79.4 | 35.0 | 77.6 | 23.0 | | | 16.6 | 195.8 |
| | | | | | $\kappa_g = 9.3$ | | $\kappa_s = 30.1$ | | $\alpha_{95} = 6.8$ | | $A_{95} = 5.5$ |
| | | A | 16/17 | 23.4 | 59.5 | 57.5 | 30.1 | | | 67.8 | 164.8 |
| | | | | | $\kappa_g = 44.5$ | | $\kappa_s = 3.5$ | | $\alpha_{95} = 23.0$ | | $A_{95} = 7.1$ |

Bedding, strike/dip with dip direction 90° clockwise from strike. Because the sites from the southern limb of the syncline are adjacent samples grouped, more than one bedding attitude was used for each site, but only one is listed here. **B**, **A**, components B and A. *n/N*, number of samples or sites used/number of samples or sites measured. *D*, *I*, declination and inclination. $\alpha_{95}$, $A_{95}$, radius of 95% confidence circle about the mean direction and pole, respectively. For component A, poles and associated $A_{95}$ were calculated from the in-situ directions. $\kappa_g$, $\kappa_s$, Fisher precision parameters before and after fold plunge and bedding tilt correction.

[a] For sites S1, S2 and N1, reversed/normal polarity samples used for component B are 3/2, 4/1, and 3/3, respectively.

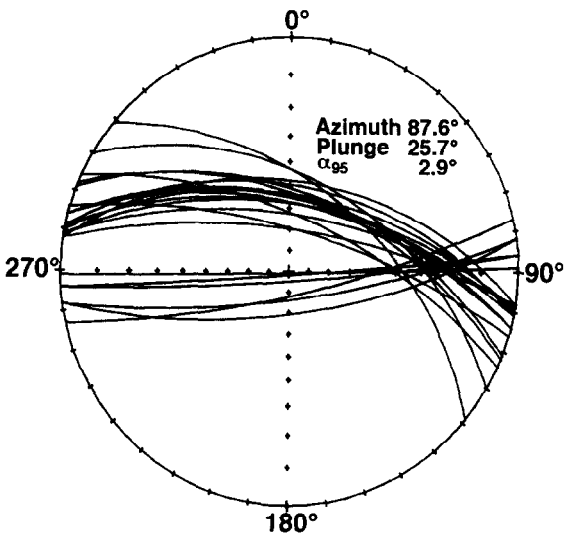



Fig. 4. Equal-angle projections of bedding planes taken from the two sampled limbs of the east-plunging syncline at Badong, Hubei. The dot represents the average fold plunge determined using an iterative method [29].

Hunan province. The structure there is a synclinorium which trends northeast in the south and bifurcates into two east-northeast-trending synclines with an intervening anticline in the north (Fig. 1). The Triassic rocks comprise the cores of the synclines and the Permian rocks crop out at the limbs. The samples were collected from the two limbs of the western branch syncline. The sampled section on the west limb, which is near a village called Hongjiaguan, about 9 km north of downtown Sangzhi, is a representative section for the Badong Formation in northern Hunan. The subdivision, lithology and fossil contents of the formation at the Hongjiaguan section correlate well with the type section in Badong, Hubei; the only difference between the two is that the sequence at Hongjiaguan is twice as thick (2118 m) as in Badong and the carbonate rocks consisting of Member 3 are more argillaceous, being mainly marlstone and argillaceous limestone [14]. Eleven sites were drilled from Member 2 at the Hongjiaguan

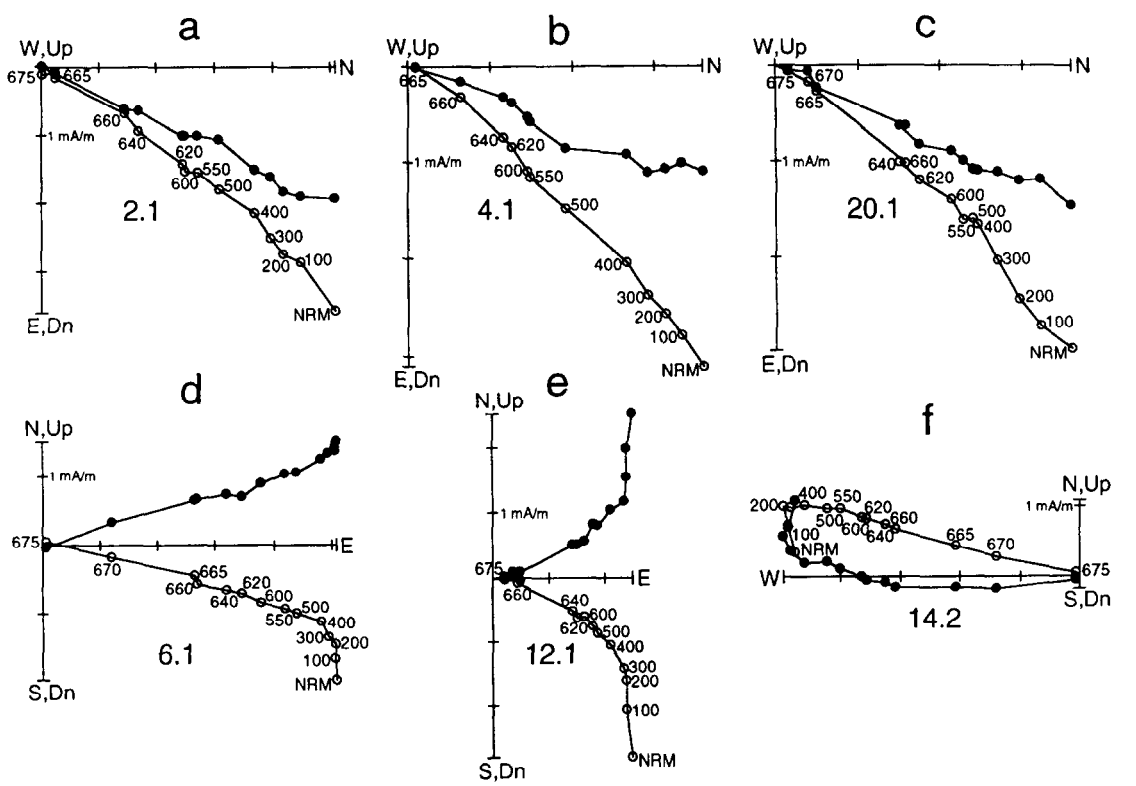



Fig. 5. Representative orthogonal plots of thermal demagnetization for the Badong Formation from Sangzhi, Hunan. Coordinates and symbols as in Fig. 2.

section and 9 sites with 2 sites extended into Member 3 were taken from the east limb near a village called Furongqiao, about 18 km northeast of downtown Sangzhi. The rocks at the sampled sites on the west and east limbs dip 119–127° southeast at 42–55° and 331–344° northwest at 37–52°, respectively.

The NRM intensities of the samples from Sangzhi range from 1.5 to 13.3 mA/m, comparable to those from Badong, Hubei. The demagnetization behavior of the samples collected from the east limb is also similar to that from Badong. Most of them possess two components: a low-temperature component was eliminated below 300°C and a ChRM component was resolved in the temperature range 400–675°C (Fig. 5d–f). The mean in-situ direction of the low-temperature component ($D/I = 6.1°/44.1°$, $\alpha_{95} = 3.8°$, $N = 54$ samples) is hardly distinguishable from the Earth's dipole field direction at Sangzhi ($D/I = 0°/48.4°$), suggesting a recent overprint origin for this component. The ChRM component, on the other hand, is of dual polarity with 5 sites normal and 4 sites reversed, and is nearly east–west directed with shallow inclinations before tilt correction, indicating that it is not a recent overprint. The samples from the west limb revealed more or less univectorial decay towards the origin of the coordinates (Fig. 5a–c). The in-situ ChRM directions of the samples from the west limb are northeasterly and moderately downward, again different from the Earth's present-day field direction. The fold test for this locality is a classic case: the site mean directions from the two limbs are well away from each other before unfolding, but converge into one group after unfolding (Fig. 6 upper). The precision parameter, $\kappa$, accordingly increases from 7.2 to 59.3 by a factor of 8.24, which is much greater than the critical value (2.15) tabulated by McElhinny [28] for a positive fold test at the 99% confidence. The tilt corrected mean direction for the Badong Formation from Sangzhi is $D/I = 63.6°/26.8°$ with $\alpha_{95} = 4.3°$, which corresponds to a pole at 29.8°N/200.0°E with $A_{95} = 4.0°$.

By applying the same techniques as used for the data from Badong, Hubei, a plunge of 47° northeast at 16.5° was determined for the sampled syncline at

Table 2
Site and formation statistics for the Badong Formation from Sangzhi (29.4°N, 110.2°E)

| Site | | Bedding | $n/N$ | In situ | Tilt corrected | | | | Pole position | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | $D$ (°) | $I$ (°) | $D$ (°) | $I$ (°) | $\alpha_{95}$ | $\kappa_s$ | Latitude (°N) | Longitude (°E) |
| 1 | 34/50 | 7/7 | 23.0 | 41.7 | 63.7 | 32.5 | 5.2 | 135.2 | 31.1 | 196.4 |
| 2 | 37/55 | 6/6 | 27.0 | 24.0 | 52.1 | 21.3 | 6.2 | 118.7 | 38.3 | 209.7 |
| 3 | 36/53 | 4/6 | 24.2 | 33.1 | 58.1 | 27.7 | 7.0 | 175.3 | 34.7 | 202.3 |
| 4 | 33/52 | 4/6 | 16.5 | 36.7 | 56.3 | 33.2 | 7.0 | 174.5 | 33.2 | 199.3 |
| 5 | 32/52 | 5/6 | 12.4 | 34.1 | 51.1 | 34.3 | 9.4 | 67.2 | 42.6 | 201.0 |
| 6 | 254/43 | 7/7 | 80.6 | 14.0 | 69.1 | 14.6 | 7.6 | 63.3 | 21.8 | 204.0 |
| 7 | 247/42 | 4/6 | 85.7 | 17.1 | 69.0 | 25.0 | 17.1 | 30.0 | 24.6 | 198.5 |
| 8 | 247/52 | 5/6 | 80.5 | 22.4 | 56.4 | 23.9 | 11.6 | 44.4 | 35.2 | 205.6 |
| 9 | 243/40 | 6/6 | 266.0 | 3.0 | 262.9 | −12.2 | 10.2 | 44.2 | −9.2 | 18.3 |
| 10 | 237/37 | 6/6 | 263.3 | −14.6 | 249.4 | −27.3 | 4.7 | 203.3 | −24.8 | 17.0 |
| 11 | 241/43 | 5/6 | 73.3 | 22.8 | 53.4 | 24.7 | 9.9 | 60.6 | 38.0 | 206.8 |
| 12 | 244/37 | 5/6 | 71.7 | 28.2 | 51.7 | 26.6 | 10.1 | 57.8 | 40.0 | 206.5 |
| 13 | 246/43 | 5/5 | 274.3 | −15.3 | 256.3 | −30.4 | 10.6 | 53.3 | −19.6 | 12.0 |
| 14 | 254/47 | 7/7 | 276.8 | −16.8 | 256.7 | −27.9 | 4.2 | 204.5 | −18.6 | 13.3 |
| 15 | 37/42 | 5/6 | 37.9 | 27.8 | 57.0 | 19.8 | 4.6 | 280.5 | 33.6 | 207.7 |
| 16 | 27/50 | 4/6 | 34.1 | 47.6 | 69.7 | 24.2 | 11.0 | 71.0 | 23.7 | 198.6 |
| 17 | 36/50 | 4/6 | 29.9 | 44.7 | 70.7 | 30.6 | 12.2 | 58.0 | 24.5 | 194.4 |
| 18 | 30/54 | 5/7 | 18.4 | 50.9 | 71.9 | 34.0 | 7.1 | 117.2 | 24.4 | 191.7 |
| 19 | 29/52 | 5/6 | 19.2 | 42.1 | 60.6 | 30.8 | 11.7 | 43.7 | 33.4 | 199.0 |
| 20 | 32/50 | 6/6 | 20.7 | 35.0 | 54.8 | 29.5 | 2.3 | 843.5 | 38.1 | 202.8 |
| Formation Mean | | 20/20 | 53.8 | 32.1 | 63.6 | 26.8 | | | 29.8 | 200.0 |
| | | | | $\kappa_g = 7.2$ | | $\kappa_s = 59.3$ | | $\alpha_{95} = 4.3$ | | $A_{95} = 4.0$ |

Explanations as in Table 1.

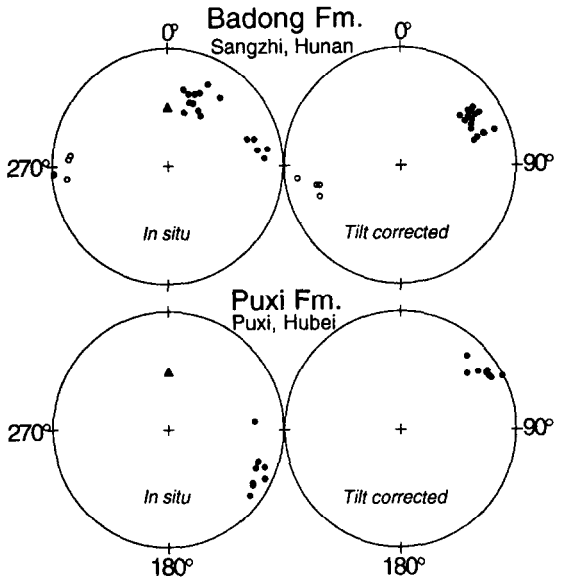



Fig. 6. Equal-area projections of site mean directions for the Badong Formation from Sangzhi, Hunan (upper) and Puxi Formation from Puxi, Hubei (lower). Symbols as in Fig. 3.

Sangzhi. A mean paleomagnetic direction for the formation of $D/I = 62.3°/27.2°$ with $\alpha_{95} = 5.6°$, $\kappa s = 35.5$ was obtained after correction for the fold plunge and bedding tilt. This is virtually identical with the results without plunge correction. Because these samples were taken from the bending section where the trend of the synclinorium changes from northeast to east-northeast, the plunge determined using the above techniques may not accurately reflect the actual attitude of the fold plunge, the results without plunge correction are preferred. Therefore, only the simple tilt-corrected data are listed in Table 2.

### *4.3. Puxi Formation from Puxi, southeastern Hubei (29.7°N, 113.9°E)*

The Puxi Formation was sampled at the type locality, Puxi city in southeastern Hubei province. The type section is along the bank of the Lushui River on the overturned south limb of a syncline, which trends east-northeast and is vergent to the north; unfortunately, it was inaccessible at the time of sampling because of flooding. Ten sites were still able to be drilled from the exposures in the nearby street and railway cuttings close to downtown Puxi. Rocks at these sites dip nearly due south at 48–58° (overturned).

The NRM intensities of the samples appear to be a bit weaker than those from the Badong Formation, with the majority in the range 1.3–5.1 mA/m. The samples from sites 1 and 4 did not yield ChRM directions at high temperatures probably due to

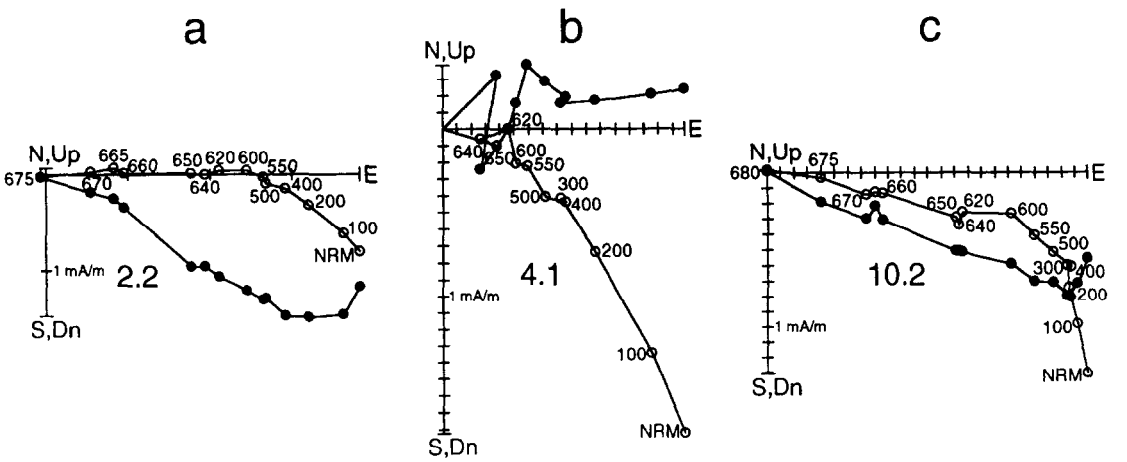



Fig. 7. Representative orthogonal plots of thermal demagnetization for the Puxi Formatiom from Puxi, Hubei. Coordinates and symbols as in Fig. 2.

weathering (Fig. 7b) and were thus discarded from further analysis. Most of the samples from other sites possess two components: a low-temperature component (A) was removed below 550°C and a ChRM component (B) was isolated above 600°C (Fig. 7a,c). In in-situ coordinates, components A and B are respectively northeasterly with moderately downward inclinations and southeasterly with shallow downward inclinations, neither of them falling close to the Earth's present-day field direction. After correction for bedding tilt, the B component becomes northeast directed and shallow downward, which is similar to the tilt-corrected ChRM direction derived from the Badong Formation at Sangzhi. The mean direction of site 5 is apparently an outlier (Fig. 6 lower part) and was excluded from the computation of the global mean. The mean direction of the Puxi Formation is $D/I = 55.1°/14.2°$ with $\alpha_{95} = 7.5°$, corresponding to a pole at 33.8°N/215.8°E with $A_{95} = 5.8°$. Site and formation statistics for Puxi are listed in Table 3.

## 5. Discussion

### 5.1. Stability of magnetization

As described earlier, except for the samples collected from the west limb of the syncline at Sangzhi, Hunan, all the other samples possess two components. The low-temperature component resolved from the east limb of the syncline at Sangzhi has been demonstrated to be a recent overprint of the Earth's magnetic field. Progressive unfolding reveals a peak value for the precision parameter, $\kappa$ (99.7), at 10% unfolding for the A component derived from Badong, Hubei, suggesting that it might be acquired at the latest stage of folding, probably during latest Jurassic or earliest Cretaceous times. The ten percent unfolded mean direction of the A component for this locality is $D/I = 24.7°/58.7°$, which corresponds to a paleopole at 68.0°N/169.7°E with $A_{95} = 5.1°$. The fact that this pole falls close to the Jura–Cretaceous segment of the APWP for the YB appears to support

Table 3
Site and formation statistics for the Puxi Formation from Puxi (29.7°N, 113.9°E)

| Site | | Bedding | n/N | | In situ | | Tilt corrected | | | | Pole position | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | D (°) | I (°) | D (°) | I (°) | $\alpha_{95}$ | $\kappa_s$ | Latitude (°N) | Longitude (°E) |
| 2 | 259/122 | 6/6 | | B | 116.3 | 7.3 | 50.6 | 26.3 | 8.0 | 70.6 | 40.9 | 211.3 |
| | | | | A | 60.1 | 43.9 | 44.7 | −34.4 | 12.2 | 31.1 | 37.2 | 192.7 |
| 3 | 259/122 | 4/6 | | B | 113.9 | 18.7 | 43.2 | 16.9 | 8.8 | 109.3 | 44.5 | 222.4 |
| | | | | A | 66.2 | 55.7 | 29.9 | −32.9 | 12.4 | 55.5 | 35.1 | 178.4 |
| 5 [a] | 263/123 | 3/5 | | B | 84.9 | 26.2 | 59.6 | −12.5 | 10.5 | 138.4 | 22.5 | 225.8 |
| | | | | A | 43.0 | 56.8 | 32.3 | −48.6 | 19.5 | 40.9 | 53.7 | 180.1 |
| 6 | 270/132 | 5/5 | | B | 109.8 | 19.1 | 62.7 | 1.1 | 8.1 | 90.4 | 23.8 | 217.7 |
| | | | | A | 51.4 | 39.9 | 75.4 | −51.8 | 21.9 | 13.1 | 43.7 | 199.9 |
| 7 | 270/132 | 6/6 | | B | 128.4 | 11.5 | 54.2 | 18.6 | 8.5 | 62.4 | 35.7 | 213.9 |
| | | | | A | 39.4 | 61.9 | 36.0 | −59.5 | 8.0 | 70.9 | 56.0 | 169.8 |
| 8 | 270/132 | 7/7 | | B | 121.7 | 15.5 | 56.7 | 11.4 | 6.8 | 79.0 | 31.6 | 216.3 |
| | | | | A | 54.3 | 59.5 | 43.0 | −52.8 | 5.5 | 122.6 | 45.0 | 175.0 |
| 9 | 270/132 | 6/6 | | B | 122.9 | 14.1 | 56.8 | 13.2 | 9.0 | 55.8 | 32.0 | 215.3 |
| | | | | A | 47.9 | 59.3 | 42.6 | −56.0 | 7.7 | 76.1 | 49.9 | 175.6 |
| 10 | 266/130 | 6/8 | | B | 111.0 | 11.3 | 60.9 | 11.0 | 11.0 | 38.0 | 27.9 | 214.1 |
| | | | | A | 37.3 | 62.3 | 30.3 | −56.6 | 7.4 | 82.7 | 57.4 | 168.3 |
| Formation Mean | | 7/10 | | B | 117.7 | 14.0 | 55.1 | 14.2 | | | 33.8 | 215.8 |
| | | | | | | $\kappa_g = 105.7$ | | $\kappa_s = 65.1$ | | $\alpha_{95} = 7.5$ | | $A_{95} = 5.8$ |
| | | | | A | 51.9 | 55.0 | 42.8 | −50.1 | | | 46.9 | 181.0 |
| | | | | | | $\kappa_g = 57.3$ | | $\kappa_s = 30.6$ | | $\alpha_{95} = 11.1$ | | $A_{95} = 8.9$ |

Explanations as in Table 1.
[a] Excluded from computation of the formation mean.

the estimated age for the A component from Badong (Fig. 8b). The same may be said about the age of the A component from Puxi (Fig. 8b).

Both the B component resolved from Badong, Hubei and the ChRM component from Sangzhi, Hunan are of dual polarity. The angular distances between the mean directions of normal and reverse polarity at these two localities are $14.4 \pm 36.2°$ and $14.8 \pm 15.1°$, respectively, indicating that their 95% confidence circles will overlap if one polarity is flipped into another in both cases. On applying a stringent reversal test developed by McFadden and McElhinny [31], however, the former case is rated as indeterminate and the latter case to be negative. This is not surprising, though, because there are fewer than five sites which are of reverse polarity and so the simulation test had to be performed in both cases. It appears that insufficient sampling of the reverse polarity made the average direction of the reversed field less accurate and uncertainties larger. Nevertheless, the presence of dual polarity in both cases indicates that enough time had lapsed during the acquisition of the ChRM.

It has been demonstrated that both the B component from Badong, Hubei and the ChRM from Sangzhi, Hunan passed McElhinny's fold test [28] at the more than 99% confidence level. Using McFadden and Jones' more stringent fold test [32], the statistic $P = 0.3010$ for the plunge and tilt corrected mean directions from the two limbs from Badong, Hubei, which is below the critical value $F_{2,28} = 0.3895$ at the 99% significance; likewise, $P = 0.1713 < F_{2,36} = 0.2915$ in the case of Sangzhi, Hunan. Therefore, the hypothesis of a common true mean direction shared by the two limbs after unfolding cannot be rejected, and the fold test is indeed significant in both cases. As regional geology indicates, the folding took place prior to the Cretaceous in western Hubei and northern Hunan; hence, we have good reason to assume that the B component from Badong and the ChRM from Sangzhi are primary.

As for the B component resolved from the Puxi Formation, the primary nature can be assumed based upon its dissimilarity to the present field direction before tilt correction and its similarity to the ChRM direction from Sangzhi after correction for bedding tilt. However, we do not regard the global mean direction derived from the Puxi Formation as robust as those from Badong and Sangzhi because it is based on a smaller number of samples taken from a monocline.

### 5.2. *Oroclinal rotations in eastern Sichuan and neighboring area*

The Middle Triassic paleomagnetic poles for the YB are very scarce. One pole based on 8 samples was reported by Chan et al [33] from the carbonate rocks of the Huaxi Formation from Guiyang. However, we have carried out an extensive sampling of the Middle Triassic rocks including the Huaxi Formation at the same locality in an attempt to refine this pole, and we were unable to reproduce their results. Instead, we found pervasive remagnetization of recent origin from these rocks (our own unpublished data). Another pole was reported by us from the Huangmaqing Formation from Nanjing [3], which was previously assigned a Late Triassic age by the regional geological survey but has been reassigned a Middle Triassic age after further investigations [13]. The original pole was based upon 23 samples with the presence of two polarities. A revisit, involving two limbs of a syncline and totaling 17 sites, has confirmed the predominant steep postfolding component; samples from only one site yielded a high-temperature component which is similar to the normal-polarity direction previously reported. No 'folding-associated' component reported by Wang and Van der Voo [9] was observed (our own unpublished data). Thus verification of the pole reported by us from Nanjing [3] is still pending. Gilder et al. [34] recently derived a pole (32.2°N, 208.7°E, $A_{95} = 7.9°$) from the Middle Triassic rocks from western Guangxi province which is traditionally considered to be part of the Huanan Fold Belt. The pole coincides with the Triassic poles derived from the undisputed part of the YB, so they concluded that Guangxi was also a part of the YB at least since the Triassic.

A number of Early Triassic paleomagnetic poles have been published for the YB (Table 4). Enkin et al. [7] noticed that these poles are streaked along a small circle centered on eastern Sichuan which indicates differential rotations between the sampled sites (Fig. 8a). In fact, the Early Triassic poles from eastern Sichuan and northern Hunan are clockwise

rotated relative to the poles from other parts of the YB (Fig. 8a) and the mean poles for these two groups (Nrs. 5 and 11 in Table 4) are significantly distinct with an angular distance of 14.9 ± 9.9°. Because the pole derived from the Lower Triassic Yinkeng Formation from the eastern part of the YB [36] (Nr. 4 in Table 4) is coincidental with the poles from northern and western Sichuan as well as northern Guizhou, eastern Sichuan and the neighboring area may have been subjected to differential clockwise rotations. One of the rotated poles (Nr. 10 in Table 4) was obtained by Dobson and Heller [38] from the Lower Triassic Daye Formation from the same (Hongjiaguan) section as our results from Sangzhi, Hunan. Comparison of this pole with the unrotated mean pole (Nr. 5 in Table 4) indicates that Sangzhi may have been rotated clockwise by 17.2 ± 9.3° relative to other parts of the YB. This seems to be supported by a bend of about 15° of the sampled syncline at Sangzhi. It is also apparent that the Middle Triassic poles we derived from the three localities of this study are discordant (Fig. 8b); the angular distance between the pole from Badong and the pole from Sangzhi is 13.7 ± 9.5° and that between Sangzhi and Puxi 14.0 ± 9.8°. This implies that Badong has been rotated clockwise since the acquisition of the remanence by 13.6 ± 8.5° with an insignificant latitudinal translation of 3.1 ± 5.5° with respect to Sangzhi, which was, in turn, possibly rotated clockwise by 10.2 ± 7.5° with a latitudinal displacement of 9.7 ± 6.3° relative to Puxi. As we indicated earlier, however, the results from Puxi are less robust than those from Badong and Sangzhi, and we would therefore attach less confidence to the Puxi results. Taking into consideration the rotation of Sangzhi inferred from the Lower Triassic data, Badong would then be rotated clockwise by 30.8 ± 12.6° relative to the other parts of the YB, while

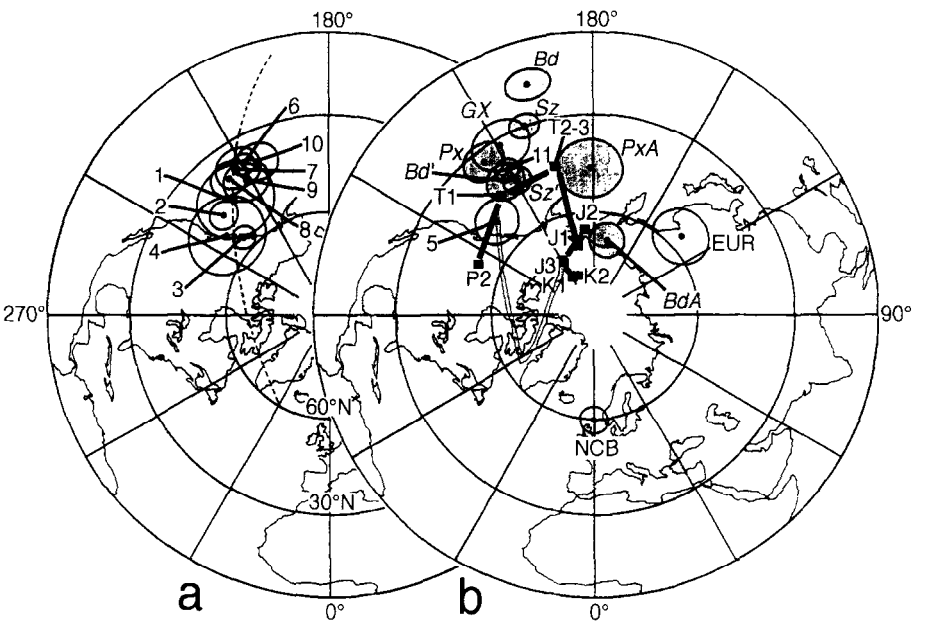



Fig. 8. (a) Early Triassic paleomagnetic poles from the YB tend to lie along a small circle (83.0 ± 5.8°, broken line) centered on eastern Sichuan (30°N, 107°E). Numbers next to the poles are the same as in Table 4. (b) The paleomagnetic poles determined from this study are shaded: *Bd* = pole calculated from the B component from Badong, Hubei; *Bd'* = pole *Bd* after correction for local rotation; *BdA* = pole calculated from the A component from Badong, Hubei after 10% unfolding; *Sz* = pole computed from the ChRM from Sangzhi, Hunan; *Sz'* = pole *Sz* after correction for local rotation; *Px* = pole computed from the B component from Puxi, Hubei; *PxA* = pole computed from the A component from Puxi. *GX* is the Middle Triassic pole derived by Gilder et al. from Guangxi [34]. Enkin et al's APWP for the YB [7] is shown with squares: *P2* = Late Permian; *T1* = Early Triassic; *T2-3* = Middle–Late Triassic; *J2* = Middle Jurassic; *J3* = Late Jurassic; *K1* = Early Cretaceous; *K2* = Late Cretaceous. Also shown are the Middle Triassic poles for the NCB [6] and Eurasia (*EUR*) [7], and the mean Early Triassic poles *5* and *11* from Table 4. The Mesozoic loop for the YB proposed by Wang et al. [10] is indicated with a shaded curve. Equal-area projection.

Table 4
Early Triassic paleomagnetic poles from the YB

| Numbers | Locality | Coordinates | $D/I$ | $N$ | $\alpha_{95}$ | Pole | $A_{95}$ | Reference |
|---|---|---|---|---|---|---|---|---|
| 1 | Tongzi, N. Guizhou | 28.6°N, 106.9°E | 40.0°/13.1° | (9) | 11.0° | 46.3°N, 219.2°E | 10.9° | [3] |
| 2 | Guangyuan, N. Sichuan | 32.4°N, 106.4°E | 36.4°/12.2° | 38 | 4.1° [a] | 47.2°N, 226.3°E | 4.1° [a] | [5] |
| 3 | Emei, W. Sichuan | 29.6°N, 103.4°E | 27.7°/15.6° | 101 | 3.3° | 56.2°N, 227.6°E | 3.3° | [35] |
| 4 | Changxing, Zhejiang | 31.0°N, 119.8°E | 35.7°/22.6° | 16 | 11.2° | 51.8°N, 232.1°E | 11.2° | [36] |
| 5 | Mean pole of 1–4 | | | | | 50.5°N, 226.1°E | 6.5° | |
| 6 | Hechuan, E. Sichuan | 29.9°N, 106.3°E | 50.4°/17.1° | 62 | 3.2° [a] | 38.5°N, 209.8°E | 3.2° [a] | [5] |
| 7 | Hechuan, E. Sichuan | 29.9°N, 106.3°E | 49.4°/14.9° | 26 | 6.1° [a] | 38.7°N, 211.8°E | 6.1° [a] | [5] |
| 8 | Changshou, E. Sichuan | 29.9°N, 107.0°E | 48.7°/16.8° | 36 | 5.1° [a] | 39.8°N, 216.2°E | 5.1° [a] | [37] |
| 9 | Nanxi, S. Sichuan | 28.9°N, 104.9°E | 44.8°/16.6° [a] | 13 | 6.2° | 43.3°N, 211.7°E | 6.2° [a] | [37] |
| 10 | Sangzhi, N. Hunan | 29.4°N, 110.2°E | 52.5°/28.5° | 15 | 5.7° | 39.8°N, 205.1°E | 5.7° | [38] |
| 11 | Mean pole of 6–10 | | | | | 40.1°N, 210.9°E | 3.4° | |

*Nr.* = Numbers denoting paleopoles in Fig. 8. $N$ = Number of samples (sites). Other explanations as in Table 1.
[a] Estimated from the data given in the original reference. Note that duplicated results from some localities are not included.

Puxi may not be significantly rotated ($7.0 \pm 11.9°$). Apparently, the amount of rotation experienced by Badong, Hubei is larger than that of Hechuan, the locality in eastern Sichuan investigated by Steiner et al. [5] ($15.7 \pm 9.1°$ to $16.7 \pm 7.3°$), as implied by the Lower Triassic data.

Three possible causes have been raised for the paleomagnetic rotations observed from eastern Sichuan and northwestern Hunan, these include regional or block rotation, thrust sheet rotation and rotation due to oroclinal bending [5,38]. Thus far, there is no geological evidence from the region which suggests that this part of the YB is a separate block, and block rotation seems to be unlikely. Differential rotations between the sampled sites observed in this study also conflict with this possibility. Local rotations caused by thrusting can not be ruled out, but rotations due to oroclinal bending are preferred for the following reasons. First, as described earlier, eastern Sichuan and neighboring area in which differential rotations have been observed are just where the prominent arcuate folds (the East Sichuan Arc and Bamianshan Arc) are developed. Geophysical surveys for oil and gas in eastern Sichuan have shown that these folds are thin skin folds confined to a shallow depth below which the folding weakens dramatically [24]. Geologically, therefore, rotations in the area are more likely associated with folds than with thrust faults. Second, it has been recognized that deviation of the paleomagnetic directions observed at some sites are consistent with the bend of the sampled folds such as at Hechuan in eastern Sichuan [5] and at Sangzhi in northwestern Hunan (this study). Third, the sampled east-trending fold at Badong in western Hubei is actually the extension of the NNE-trending folds in eastern Sichuan (Fig. 1); the larger amount of rotation observed at Badong appears to correlate with the more easterly structure trend when compared with the smaller amount of rotation and more northerly trend of the fold observed at Hechuan. Since there are only two localities studied along the East Sichuan Arc at present (Hechuan and Badong), more intensive paleomagnetic investigations are required in order to conclusively test this oroclinal bending interpretation. If this turns out to be true, the oroclinal bending in eastern Sichuan and neighboring area would have nothing to do with the collision of India with Tibet or with the collision of the NCB with YB; instead, it might be due to overthrusting from the southeast sometime during Late Jurassic or Cretaceous times.

## 6. Post-Middle Triassic suturing between the YB and NCB

Reliable Middle Triassic paleomagnetic poles are also scarce for the NCB. The only good pole for this period was reported by Yang et al. [6] from the Ordos basin with 17 sites and positive fold and reversal tests. It is immediately apparent from Fig. 8b that the YB, NCB and Eurasia had not finally sutured together by Middle Triassic time because the Middle Triassic poles for these three blocks are still

far apart even after the poles for the YB are corrected for the local rotations. The observed Middle Triassic paleolatitudes for Badong and Sangzhi are $12.0 \pm 3.8°$N and $14.2 \pm 2.5°$N; the paleolatitudes for these two localities expected from the unrotated Early Triassic mean pole for the YB (Nr. 5 in Table 4) are $9.3 \pm 6.5°$N and $7.9 \pm 6.5°$N, respectively. About 2.7–6.3° northward latitudinal change for these localities had taken place during this period, which converts to a plate motion rate of 5.5–12.7 mm/yr assuming the time interval from the Triassic/Permian boundary to Anisian/Ladinian Stage boundary being 5.5 Ma [39]. This is a pretty fast rate, suggesting that subduction of the Qinling Ocean between the YB and NCB was proceeding vigorously during this time. On the other hand, the paleolatitudes for Sangzhi and Badong predicted from Yang et al.'s Middle Triassic pole are $15.8 \pm 4.0°$N and $17.1 \pm 4.0°$N, and the declinations for these two localities predicted from the same pole are $330.6 \pm 4.2°$ and $330.5 \pm 4.2°$, respectively. The differences in paleolatitude between the predicted and observed for Sangzhi and Badong are thus $1.6 \pm 4.7°$ and $5.1 \pm 5.5°$, respectively, which are insignificant within the error limits of paleomagnetism. But the differences in declination between the observed and predicted for these two localities are substantial. For instance, the difference for Sangzhi is still $75.8 \pm 11.3°$ after correction for the local rotation. Such difference is comparable to that during Late Permian time (67°, see [4]). The implication is that by the Middle Triassic the eastern end of the YB nearly docked against the NCB, and that the large discordance in paleoorientation between the two blocks had to be overcome, possibly through opposite rotations, before they finally reached their present configuration. Therefore, present data suggest that the scissor-like suturing between the YB and NCB postulated by Zhao and Coe [4] is still the most reasonable model. Perhaps, that the occurrence of high-pressure metamorphism of the crustal rocks of the YB origin has only been found in the eastern part of the Qinling Fold Belt (Dabieshan and Su-Lu areas) is due to the earlier collision and subsequent intracontinental thrusting in this part of the collision zone. Note also that the Middle Triassic pole (*GX* in Fig. 8b) obtained by Gilder et al. from Guangxi [34] overlaps with the pole from Puxi and the rotation-corrected poles from Badong and Sangzhi (*Bd'* and *Sz'* in Fig. 8b), and this suggests that Guangxi may indeed be part of the YB since the Middle Triassic.

Finally, as can be seen from Fig. 8b, the Middle Triassic poles derived from this study (*Px*, *Bd'* and *Sz'*) as well as the pole from Guangxi (*GX*) are all located to the east of the unrotated Early Triassic mean pole (5), and they basically follow the APWP for the YB determined by Enkin et al. [7]. The direction of the polar wander for the YB for this period is apparently opposite to that of the Mesozoic loop proposed by Wang et al. [10], although reliable paleomagnetic data from the Jurassic will be the key to the resolution of this controversy.

## 7. Conclusions

(1) Prefolding, presumably primary, ChRM has been resolved from the redbeds from the Middle Triassic Badong Formation from Badong, Hubei and Sangzhi, Hunan. Similar ChRM directions have also been obtained from the coeval Puxi Formation from southeastern Hubei. The data indicate that Badong has been rotated clockwise by $13.6 \pm 8.5°$ relative to Sangzhi since the acquisition of the remanence which, in turn, has possibly been rotated in the same sense relative to the other parts of the YB except for eastern Sichuan. The latter inference is based upon the change of local structure trend and published Lower Triassic paleomagnetic data for the YB. Together, the Lower and Middle Triassic data from the YB appear to support that the cover strata in eastern Sichuan and the border area between Sichuan, Guizhou, Hubei and Hunan have suffered differential rotations, probably due to oroclinal bending [5].

(2) Comparison with the coeval paleopole from the NCB indicates that the final amalgamation between the YB and NCB was post-Middle Triassic.

(3) The procedures used in this study for determining and correcting a fold plunge on paleomagnetic directions are useful, especially when the sampled fold limbs dip steeply.

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